Review: Slope as rate of change

1 Determine whether the following is a correct description of slope:

a $\dfrac{\text{Horizontal rise}}{\text{Vertical run}}$ **b** $\dfrac{\text{Vertical rise}}{\text{Horizontal run}}$ **c** $\dfrac{\text{Horizontal run}}{\text{Vertical rise}}$ **d** $\dfrac{\text{Vertical run}}{\text{Horizontal rise}}$

2 State the term used to the change in the horizontal distance along a line.

3 State the definiton of the slope of a line.

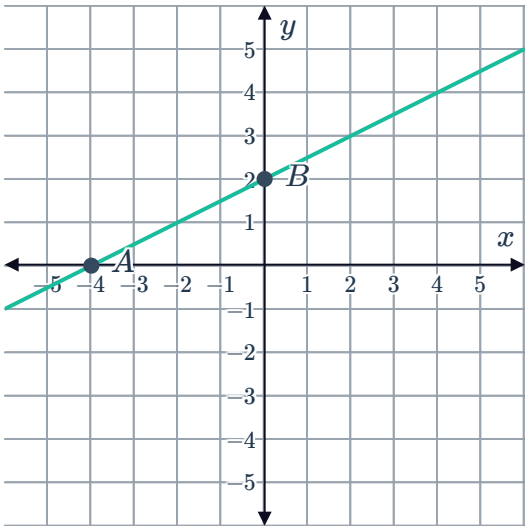
4 For each of the following segments containing A and B:

i Find the rise going from A and B.

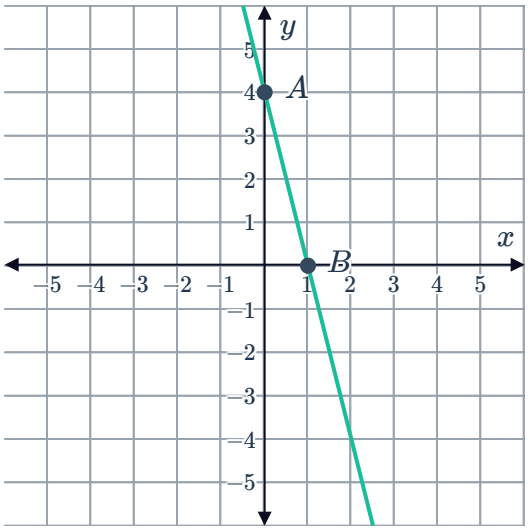
ii Find the run going from A and B.

iii Find the slope of the $\overline{AB}$.

a $A\,(-4,0)$ and $B\,(0,2)$



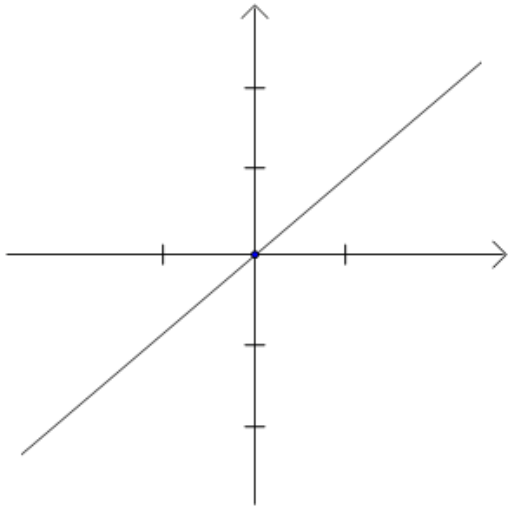
b $A\,(0,4)$ and $B\,(1,0)$



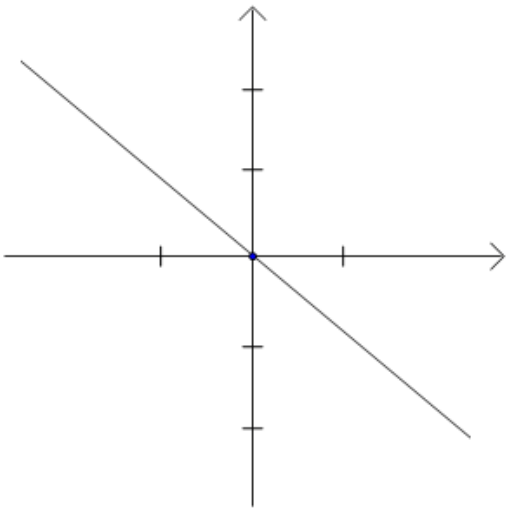
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5 Identify what kind of slope the following lines have.

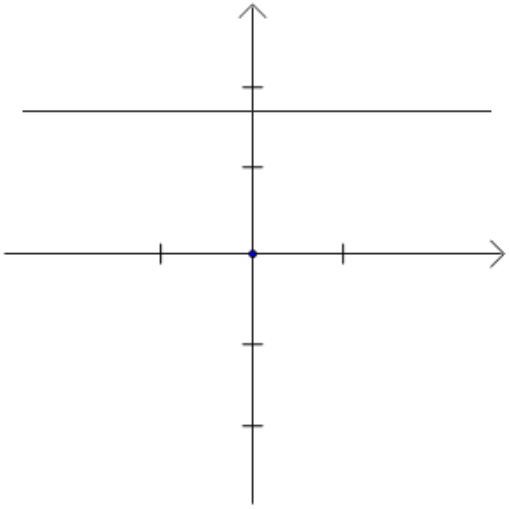
a



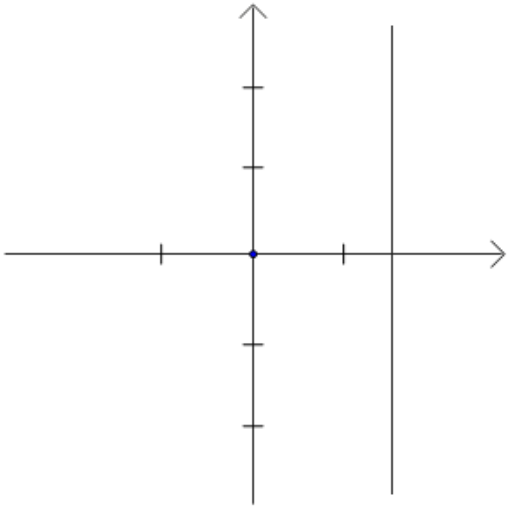
b



c



d

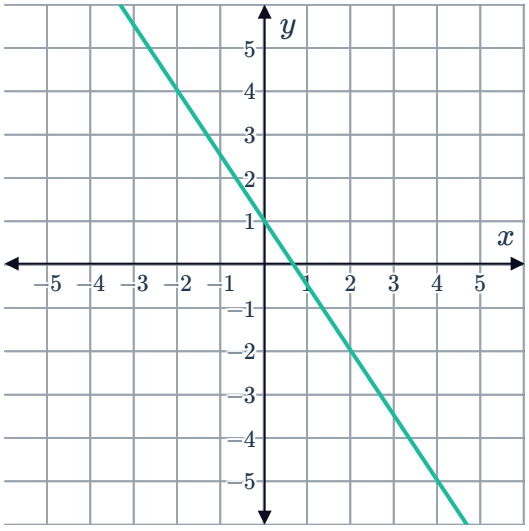


6 A line crosses the x-axis at 7, and crosses the y-axis at -6.
Is the slope of the line positive or negative?

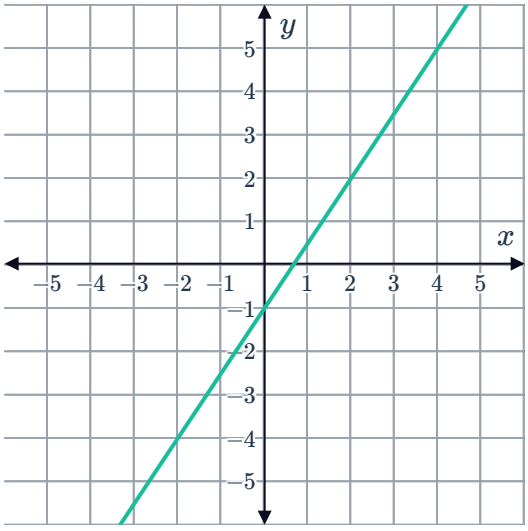
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7 The slope of a line is $\frac{3}{2}$, and when $x = 0$, $y = -1$.
Determine whether the following could be the graph of the line:

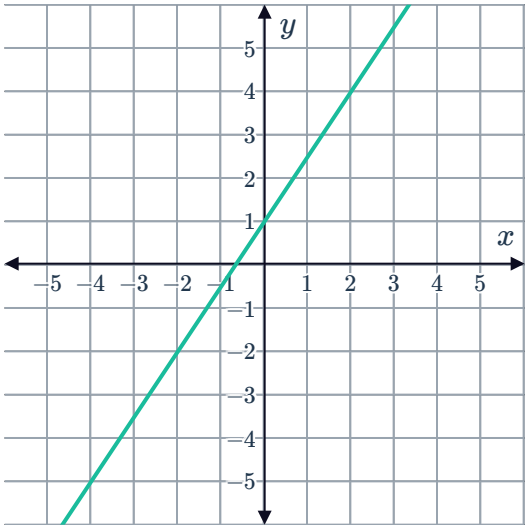
a



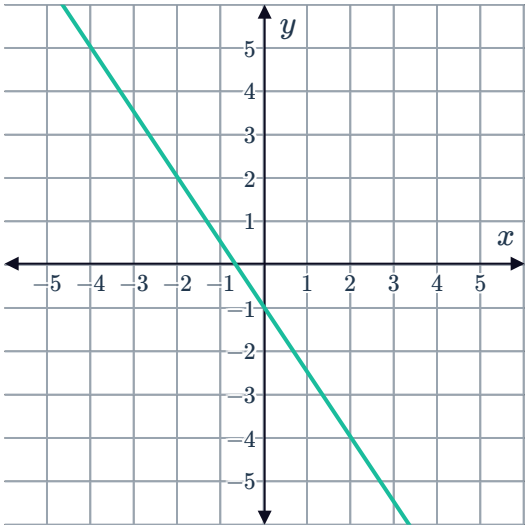
b



c



d



8 A staircase is to be built so that its maximum steepness is 0.6.
If each step goes in $22\ \text{cm}$, solve for the maximum height, in centimeters, it can rise vertically.

9 A paratrooper falls to the ground along a diagonal line. His fall begins $1157\ \text{m}$ above the ground, and the line he follows has a slope of 1.3. That is, he falls $1.3\ \text{m}$ vertically for every $1\ \text{m}$ he moves across horizontally.
How far horizontally across the ground does he land from his initial position in the sky?

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10 Gas costs a certain amount per gallon. The following table shows the cost in dollars of various amounts of gas.

Number of gallons (x)	0	10	20	30	40
Cost of gas (y)	0	12.70	25.40	38.10	50.80

a Write an equation linking the number of gallons of gas pumped (x) and the cost of the pgasetrol (y).

b How much does gas cost per gallon?

c How much would 73 gallons of gas cost at this unit price?

d In the equation, $y = 1.27x$, what does 1.27 represent?

11 A diver starts at the surface of the water and begins to descend below the surface at a constant rate. The following table shows the depth of the diver over 5 minutes.

Number of minutes passed (x)	0	1	2	3	4
Depth of diver in meters (y)	0	1.4	2.8	4.2	5.6

a What is the increase in depth each minute?

b Write an equation for the relationship between the number of minutes passed (x) and the depth (y) of the diver.

c In the equation, $y = 1.4x$, what does 1.4 represent?

d At what depth would the diver be after 6 minutes?

e How long does the diver take to reach 12.6 m beneath the surface?

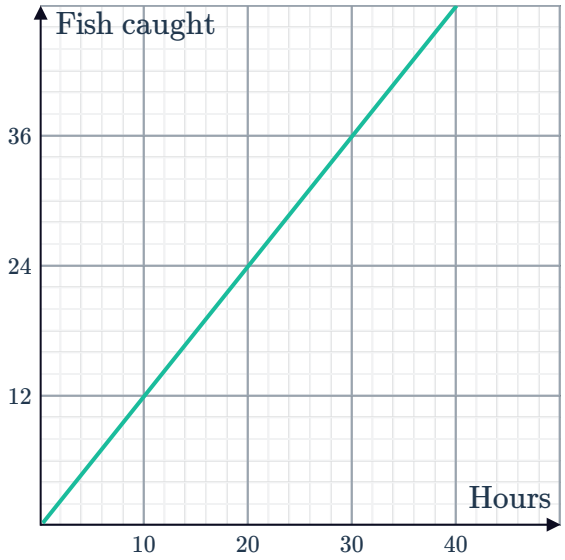
12 A dam used to supply water to the neighboring town had the following data recorded for its volume over a number of months.

Month	1	2	3	4
Volume (billions of liters)	134	117	100	83

At what rate is the volume of water in the dam changing?

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13 Consider the proportional relationship shown in the graph, showing the number of fish caught over time.



a What is the unit rate of this relationship in fraction?

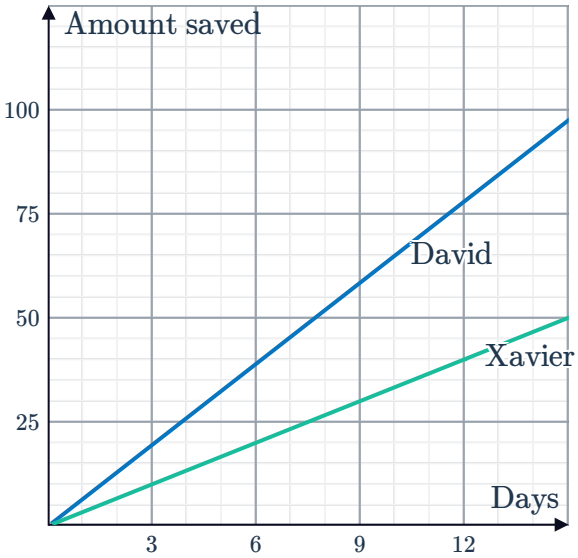
b Interpret your answer from part (a).

c What feature of the straight line graph does the rate of change relate to?

14 David and Xavier are saving money for a vacation. David knows that he can represent his savings over time using the equation $y = \frac{13}{2}x$, where x represents the number of days and y represents the savings.

a Xavier saves $\$10$ every 3 days. Complete the table showing Xavier's savings over time.

Days	▯	6	9	12
Dollars saved	10	20	30	▯



b Comparing the graphs of each person's savings, how can you tell who is saving more per day?

c Using the graph, who has the greater rate of savings?

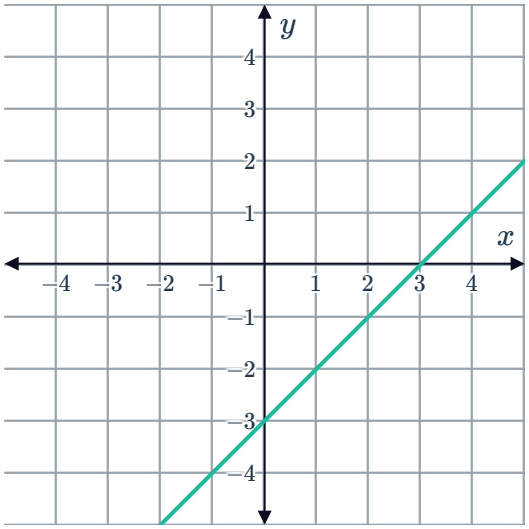
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Additional questions

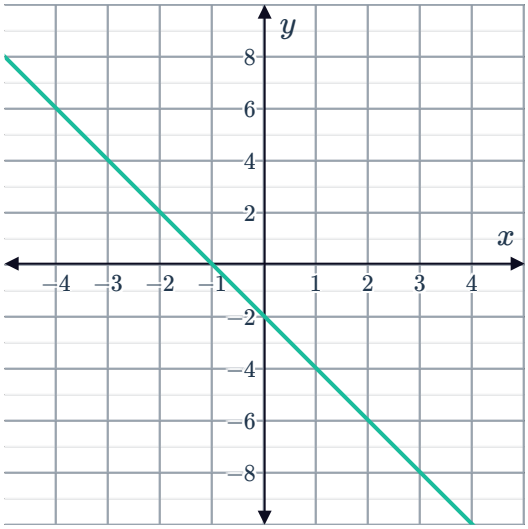
15 For the following functions:

- **i** State whether the function is increasing, decreasing, or constant.
- **ii** State the rate of change.

a



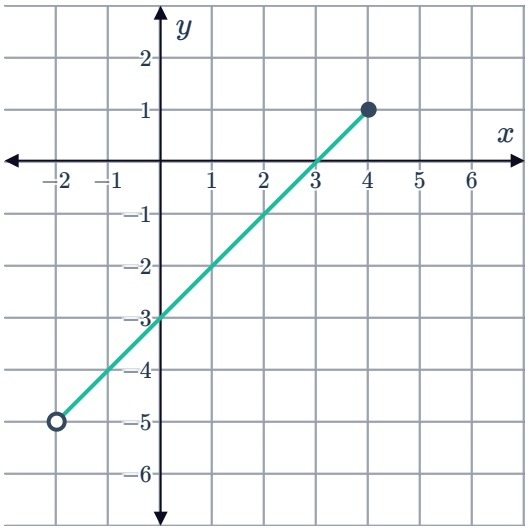
b



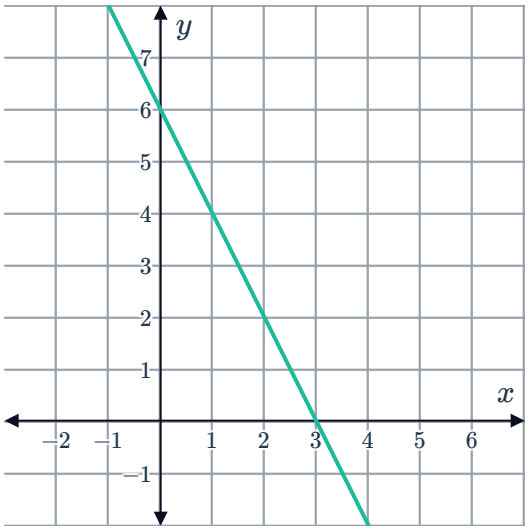
16 For each of the following graphs:

- **i** State the domain.
- **ii** State the range.
- **iii** State the x- and y-intercepts.
- **iv** State the rate of change.

a

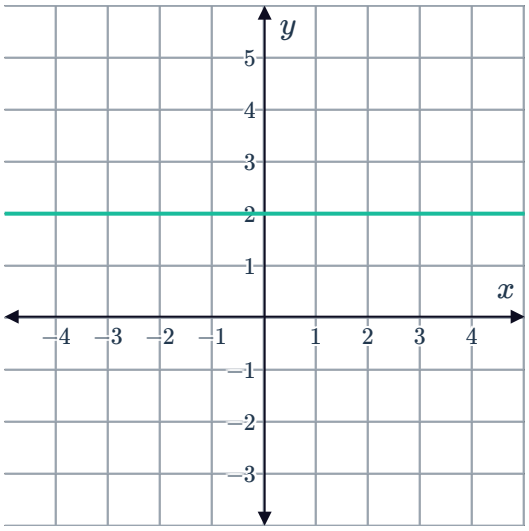


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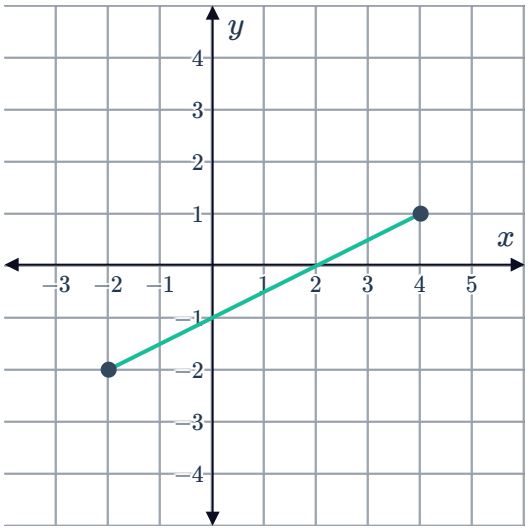


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c



d

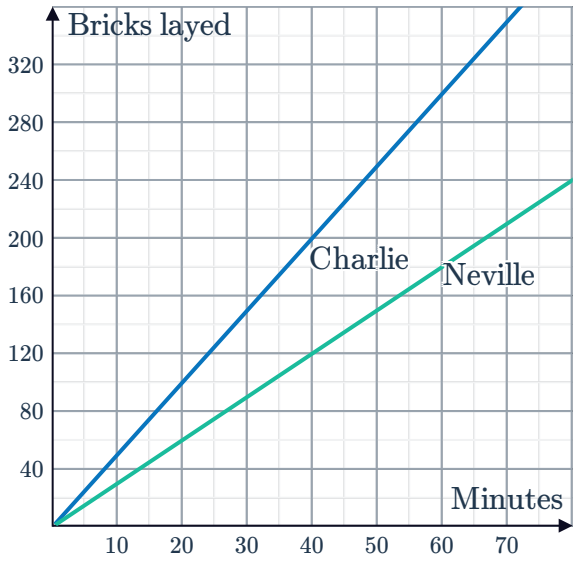


17 Two construction workers are competing to see who can lay the most bricks in one hour. The graph shows the number of bricks layed and the time, in minutes.

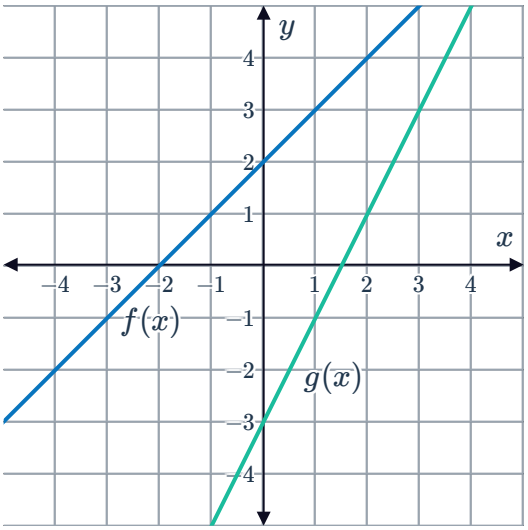
a Interpret the rate of change of the given lines.

b Determine who can lay more bricks in 60 minutes.

c Explain why the y-intercept of both lines is 0.

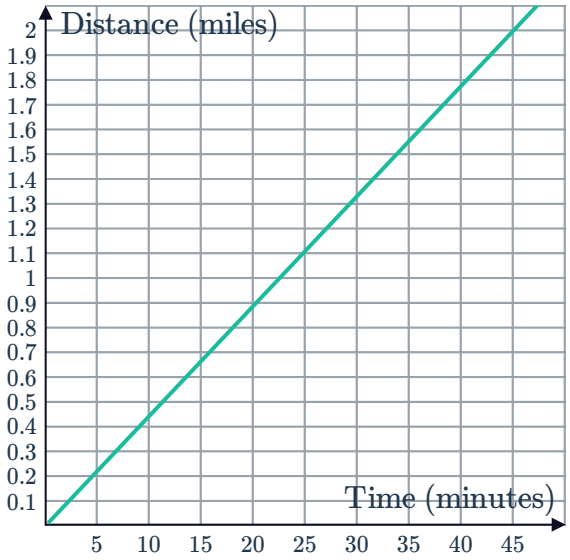


18 Alicia has concluded that the rate of change is greater in $f(x)$ than $g(x)$ because it is higher on the given graph.
Explain and correct her error.



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19 The following graph shows the distance Natalia swam during a recent open water swim:



a What is the slope of the line?

b What does the slope of the line represent in terms of the situation?

20 The following graph shows the number of batches of cookies produced in a factory:



a What is the slope of the line?

b What does the slope of the line represent in terms of the situation?

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